

STATISTICAL ANALYSIS REPORT

1. Introduction

This report analyzes the dataset to understand patterns in customer orders, revenue generation, and relationships between key variables. The goal is to use statistical methods to support business decisions with data-driven insights.

2. Descriptive Statistics -Order values

The dataset was analysed using basic statistical measures to understand the distribution of order value.

The mean is approximately 2725 and median is approximately 1052.58 which is lower than the mean. This suggests that the data is slightly skewed, with some higher value pulling the average upward, while the standard deviation is 1162, showing that there is a moderate level of variation in order value.

A histogram was also created to visualize the distribution of order values.

From this analysis, it is clear that order values vary across customers, indicating differences in purchasing behavior.

Metrics	Values
Total transactions	5000
Mean order value	2725.01
Median order value	1052.58
Standard deviation	1162.457

3. Hypothesis Testing

● Test 1: Customer Spending Behavior

The first test was conducted to check whether email customers spend more than non-email customers.

- Null Hypothesis (H_0): There is no difference in spending

- Alternative Hypothesis (H_1): Email customers spend more

The dataset did not contain a direct indicator to identify email and non-email customers. Therefore, a proxy classification was created using the customer segment variable. Customers categorized as "VIP" and "Regular" were assumed to be email users, while the remaining customers were considered non-email users.

T- Test Result

P -value = 0.8128



Since the p-value (0.812) is greater than the significant level of 0.05, the result is not statistically significant.

Conclusion:

There is no statistically significant difference in average spending between email and non – email customer. Although email users show slightly higher average revenue, the difference is not meaningful based on the t-test results.

● Test 2: A/B Testing (Checkout Conversion)

The second test evaluated whether a new checkout process improves conversion rates.

- Null Hypothesis (H_0): Both versions perform equally

- Alternative Hypothesis (H_1): New checkout performs better

Metric	Control (old)	Variant(new)
Users	961	1040
Conversions	153	278
Conversion rate	16%	27%
1	z score	5.8767
2	p value	0.00000000419

Conclusion:

Since the p value (0.00000000419) is far below the significance level of 0.05, the result is highly statistically significant. The new checkout process improves conversion rate, increasing from 16% to 27%. This percentage supports adopting the new checkout.

4. Correlation and Regression

◆ Correlation

A correlation analysis was performed to understand the relationship between price and sales.

The correlation value was approximately 0.01, which is very close to zero.

This indicates a very weak relationship between price and sales. In simple terms, changes in price do not significantly affect sales in this dataset.

◆ Linear Regression Analysis

Objective :

The purpose of this analysis was to examine how the number of orders impacts revenue and to build a predictive model.

Results :

A linear regression model was created using number of orders as the independent variable and revenue as the dependent variable.

◆ The R-squared value is approximately 0.98

Interpretation :

This means that around 98% of the variation in revenue can be explained by the number of orders.

This is a very strong relationship, indicating that as the number of orders increases, revenue also increases significantly.

The scatter plot with a trendline further confirms this positive relationship.

Regression Equation :

«Revenue = (Slope × Number of Orders) + Intercept»

This equation can be used to predict future revenue based on expected order volume.

6. Key Insights

- **There is a strong positive relationship between orders and revenue**
- **Price does not significantly impact sales**
- **Order volume is the main driver of revenue**
- **The regression model is highly reliable for prediction**

7. Conclusion

Overall, the analysis shows that business performance is largely driven by the number of orders. The strong regression results indicate that increasing order volume will directly increase revenue. Additionally, pricing alone does not seem to influence sales significantly, suggesting that other factors such as customer engagement or marketing may play a bigger role.

8. Recommendations

- **Focus on strategies to increase the number of orders**
- **Improve customer engagement and retention**
- **Use the regression model for forecasting future revenue**
- **Perform deeper analysis on customer segments for better targeting**

End of Report